Automorphisms groups of oligomorphic groups

Philipp Schlicht, University of Siena

XXIX Incontro di Logica AILA, Roma, 17 September 2026

We study oligomorphic permutation groups on $\mathbb{N}$ and other classes non-Archimedean Polish groups (subgroups of S_∞).

Problem

What can we say about the outer automorphism group

$$\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)?$$

This is joint work with Andre Nies (University of Auckland)

References

- Nies, Schlicht: Automorphism groups of non-Archimedean groups accepted for Fundamenta Mathematicae in 2026

Topology of $\mathrm{Aut}(G)$

Let H be a closed subgroup of S_∞ (denoted $H \leq_c S_\infty$).

By $N_{S_\infty}(H)$ we denote the normaliser, and by $C_{S_\infty}(H)$ the centraliser, of H in S_∞. Both are closed subgroups of S_∞.

Theorem

Suppose G is a Polish group with a countable neighbourhood basis $\mathcal{S}_G$ of the neutral element consisting of open subgroups, such that $\mathcal{S}_G$ is invariant under the action of $\mathrm{Aut}(G)$ on the open subgroups.

(i) There is a group $\hat{G} \leq_c S_\infty$ and a topological isomorphism $\Theta\colon G \to \hat{G}$ such that

$$N_{S_\infty}(\hat{G})/C_{S_\infty}(\hat{G}) \cong \mathrm{Aut}(G)$$

via sending an $\alpha \in N_{S_\infty}(\hat{G})$ to its conjugation action on G.

Using this isomorphism, $\mathrm{Aut}(G)$ can be topologised as a non-Archimedean group such that its action on G is continuous.

Topology of $\mathrm{Aut}(G)$

(ii) A neighbourhood basis of the identity for this topology on $\mathrm{Aut}(G)$ is given by the subgroups of the form

$$\{\Phi \in \mathrm{Aut}(G) \colon \bigwedge_{i=1}^{n} [\Phi(A_i) = A_i]\}, \qquad (*)$$

where $A_1, \ldots, A_n$ are cosets of subgroups in $\mathcal{S}_G$.

(iii) This topology on $\mathrm{Aut}(G)$ is the unique Polish topology that makes the action of $\mathrm{Aut}(G)$ on G continuous.

The proof idea is to identify an element of $\mathrm{Aut}(G)$ with its action on the left cosets of subgroups in $\mathcal{S}_G$.

Two opposite classes

Which classes groups satisfy the above condition?

- Oligomorphic:

  for each k, the action on $\mathbb{N}^k$ has only finitely many orbits

  These are the automorphism groups of ω-categorical structures with domain $\mathbb{N}$.

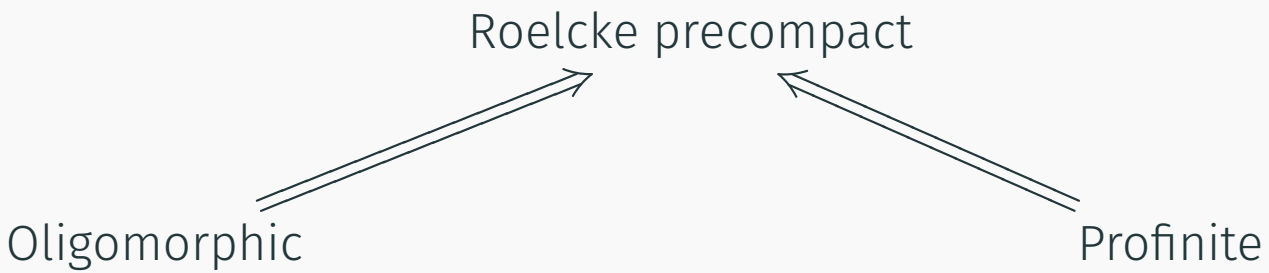
- Profinite:

  each orbit of the action on $\mathbb{N}$ is finite.

  These are the compact subgroups of S_∞ and up to isomorphism, the inverse limits of finite groups.

A Borel superclass

A closed subgroup G of S_∞ is Roelcke precompact if each open subgroup U of G is large in the sense that there is finite set $F \subseteq G$ such that $UFU = G$.



Background on oligomorphic and Roelcke precompact groups:

- Tsankov: Unitary representations of oligomorphic groups 2012

Oligomorphic groups

- A closed subgroup G of S_∞ is called oligomorphic if for each k, the action of G on $\mathbb{N}^k$ has only finitely many orbits.
- For instance, $\mathrm{Aut}(R)$ and $\mathrm{Aut}(\mathbb{Q}, <)$ are oligomorphic.
- This is the opposite of profinite, where each orbit is finite.
- Intuitively, oligomorphic groups are big, profinite groups are small.
- Unlike for profinite groups, G being oligomorphic depends on the way G is embedded into S_∞.

Oligomorphic groups as automorphism groups

Fact

$G \leq_c S_\infty$ is oligomorphic $\iff$ G is the automorphism group of an ω-categorical structure S with domain $\mathbb{N}$.

Proof.

$\Leftarrow$: this follows from the Ryll-Nardzewski Theorem.

$\Rightarrow$: $G = \mathrm{Aut}(S)$ where S is the structure with a k-ary relation symbol for each orbit of G on $\mathbb{N}^k$. $\square$

Roelcke precompactness

A closed subgroup G of S_∞ is called Roelcke precompact if for each open subgroup U of G

there is a finite set $F \subseteq G$ such that $UFU = G$.

This condition is Borel because it suffices to check it for the basic open subgroups $U_n = \{\rho \in G\colon\ \forall i < n\, [\rho(i) = i]\}$; further, we can pick F from a countable dense set predetermined from G in a Borel way.

Fact. G Roelcke precompact $\Rightarrow$

G has only countably many open subgroups.

Proof. Each open subgroup U contains a basic open subgroup U_n.

U_n has finitely many double cosets, and U is the union of some of them.

In fact, from G we can in a Borel way determine a listing $A_0, A_1, \ldots$ without repetition of all open cosets.

Roelcke precompactness

Fact. (Tsankov) Every oligomorphic group is Roelcke precompact

Fact. (Tsankov) Roelcke precompact Polish groups are precisely the inverse limits of oligomorphic groups.

Fact. (Ben Yaacov, Tsankov) Roelcke precompact Polish groups are precisely the automorphism groups of ω-categorical metric structures.

Fact. The open subgroups of an oligomorphic group $G \curvearrowright \mathbb{N}$ are exactly the stabilizers $G_{\vec{a}/E}$ of imaginaries.

An imaginary is an equivalence class of an $\emptyset$-definable equivalence relation.

Much more can be found in:

- Evans: Homogeneous structures, ω-categoricity and amalgamation constructions
  Notes on a Minicourse given at HIM, Bonn, September 2013

Local Roelcke precompactness

Definition (Zielinski)

A Polish group G is called locally Roelcke precompact if some open subgroup $U \leq G$ is Roelcke precompact.

Fact. Every totally disconnected locally compact (t.d.l.c.) group is locally Roelcke precompact.

Proposition (Zielinski 2018)

The group $\mathrm{Aut}(T_\infty)$ of automorphisms of an infinitely branching (unrooted) tree is loc. Roelcke precompact, but neither Roelcke precompact not locally compact.

Outer automorphism groups

An automorphism π of a group G is called **inner** if it is a conjugation $\pi_h(g) = h^{-1}gh$ by an element of g.

The **outer automorphism** group $\mathrm{Out}(G)$ of a group G is defined as the quotient $\mathrm{Aut}(G)/\mathrm{Inn}(G)$.

There is an informal connection between the outer automorphism group $\mathrm{Aut}(G)/\mathrm{Inn}(G)$ of an oligomorphic group G and the complexity of the isomorphism problem

These problems are connected by the existence of non-trivial outer automorphisms.

Why is $\mathrm{Out}(G)$ related to the isomorphism problem?

The study of $\mathrm{Out}(G)$ is informally related to the (top.) isomorphism problem for the class of groups G. Both are related to the existence of non-inner automorphisms.

The conjugacy relation for oligomorphic groups is smooth. To see this,

- Given a closed subgroup G of S_∞, let V_G be the corresponding orbit equivalence structure: for each $k > 0$ introduce a $2k$-ary relation that holds for two k-tuples of distinct elements if they are in the same orbit of $\mathbb{N}^k$.
- V_G is ω-categorical.
- One checks that for oligomorphic groups G, H

$$G \text{ and } H \text{ are conjugate in } S_\infty \iff V_G \cong V_H.$$

- Isomorphism of ω-categorical structures M, N for the same language is smooth because $M \cong N \iff \mathrm{Th}(M) = \mathrm{Th}(N)$.

Outer automorphism groups

A structure M has **no algebraicity** if for any finite subset $F \subseteq M$, the algebraic closure $\mathrm{acl}(F)$ of F equals F.

Theorem (Nies, Paolini 2025)

For the class of oligomorphic groups G without algebraicity:

1. *The (top.) isomorphism problem is smooth.*
2. $\mathrm{Out}(G)$ *is profinite.*

Understanding the outer automorphism group

An *interpretation* of a structure A in a structure B is a representation of A as a definable set of k-tuples in B quotiented by a definable (without parameters) equivalence relation.

The relations and functions of A are also represented in a definable way.

An interpretation is given by a scheme Γ of formulas.

A theory S is *interpretable* in a theory T if for any structure B with $B \models T$, there is some A with $A \models S$ that can be interpreted in B.

Examples:

- The theory of algebraically closed fields of characteristic 0 is interpretable in the theory of real closed fields.

Interpretability

An *interpretation* of a structure A in a structure B is a representation of A as a definable set of k-tuples in B quotiented by a definable (without parameters) equivalence relation.

The relations and functions of A are also represented in a definable way.

An interpretation is given by a scheme Γ of formulas.

Examples:

- Let M be a countably infinite structure in the empty language. Let N be a countably infinite structure with an equivalence relation with all classes of size 2.

  Then M can be interpreted in N. Conversely, N can be interpreted in M: quotient M^3 by the equivalence relation: $(a, b, c) \sim (a', b', c') \Leftrightarrow ((a = b) \wedge (a' = b') \wedge (c = c')) \vee ((a \neq b) \wedge (a' \neq b') \wedge (c = c'))$.

Bi-interpretability

Structures A, B are *bi-interpretable* if there are

- interpretations Γ, of A in B, and Δ, of B in A (all formulas without parameters)
- isomorphisms $\gamma\colon A \cong \Gamma(B)$, $\delta\colon B \cong \Delta(A)$ such that $\delta \circ \gamma$ is definable in A (without parameters) and similarly for $\gamma \circ \delta$.

(Note that $\hat{A} = \Gamma(\Delta(A))$ consists of equivalence classes of tuples from A.)

Coquand[1] showed that for ω-categorical A, B we have

$\mathrm{Aut}(A) \cong \mathrm{Aut}(B) \iff A$ and B are bi-interpretable.

[1] see Ahlbrandt, Ziegler 1986 or Evans' 2013 Bonn lecture notes

Bi-interpretability

Structures A, B are *bi-interpretable* if there are

- interpretations Γ, of A in B, and Δ, of B in A (all formulas without parameters)
- isomorphisms $\gamma\colon A \cong \Gamma(B)$, $\delta\colon B \cong \Delta(A)$ such that $\delta \circ \gamma$ is definable in A (without parameters) and similarly for $\gamma \circ \delta$.

Let A, B be ω-categorical structures.

$\mathrm{Aut}(A) \cong \mathrm{Aut}(B) \iff A$ and B are bi-interpretable.

Example:

- Let M be a countably infinite structure in the empty language. Let N be a countably infinite structure with an equivalence relation with all classes of size 2.

  Then M can be interpreted in N and conversely.

Since $\mathrm{Aut}(M) \ncong \mathrm{Aut}(N)$, M, N are not bi-interpretable.

Outer automorphism groups

We showed that the outer automorphism group of an oligomorphic group G is always locally compact in a natural topology on it. This was also shown by Paolini and Nies by different methods.

The idea is that for ω-categorical theories T and U, the set of interpretations $\alpha\colon T \to U$ forms the path space $\mathrm{Int}(T, U)$ of a tree.

- Only the root can be infinitely branching. This first branching encodes the triples $c = \langle k, \phi_D, \phi_E \rangle$ that T allows.
- Each level below is dedicated to a relation symbol of L_U: it tells us that $Rx_1 \dots x_n$ is mapped to a formula of L_T that is invariant under E. Extra conditions ask for formulas in T to hold in the interpretation in U.

This set has a natural locally compact, totally disconnected topology. This provides a quotient topology on the set of bi-interpretations.

Outer automorphism groups

Definition

We call an interpretation $\alpha\colon D/E \to \mathbb{N}$ *full* if for any $m \in \mathbb{N}$, α^m sends definable subsets of $(D/E)^m$ to definable subsets of $\mathbb{N}^m$. In other words, α^m sends E^m-invariant definable subsets on D^m to definable subsets of $\mathbb{N}^m$.

Lemma

The following are equivalent for interpretations α and β of dimensions k and l, respectively.

1. (α, β) *is a bi-interpretation.*
2. $\beta\alpha^l$ *is definable and* α *is full.*

In this case $\alpha\beta^k = \alpha^{kl}[(\beta\alpha^l)^k)]$.

Outer automorphism groups

Remark

The assumption of fullness is important. $\beta\alpha^l$ being definable does not imply that (α, β) is a bi-interpretation.

For example, the partial orders $(\mathbb{Q}, \leq)$ and $(\mathbb{Q}^2, \leq^*)$ with $(m_0, n_0) \leq^* (m_1, n_1)$ if $m_0 < m_1$ and $n_0 = n_1 = 0$, or $m_0 = m_1$ and $n_0 < n_1$, are mutually interpretable with interpretations α and β such that $\beta\alpha$ is definable, but $\alpha\beta$ is not. Note that both structures are ω-categorical.

Outer automorphism groups

Suppose M is ω-categorical. Let $\mathrm{Int}(M)$ denote the set of **self-interpretations** of M in M.

Definition (Ahlbrandt, Ziegler)

$\alpha, \beta \in \mathrm{Int}(M)$ are called *homotopic* ($\alpha \sim \beta$) if there is a definable bijection $f\colon D_0/E_0 \to D_1/E_1$ such that $\alpha = \beta f$.

Equivalently, the relation $\alpha(x) = \beta(x)$ is definable in M.

Definition (similarity for models)

We call $\alpha, \beta \in \mathrm{Int}(M)$ *similar* ($\alpha \approx \beta$) if there is a definable bijection $f\colon D_0/E_0 \to D_1/E_1$ such that $f^m(\alpha^m)^{-1}(A) = (\beta^m)^{-1}(A)$ for all definable subsets A of $\mathbb{N}^m$ and any $m \in \mathbb{N}$.

We can define **similarity for theories** analogously.

Lemma

Suppose that $\alpha, \beta \in \mathrm{Int}(M)$.

1. $\alpha\pi \approx \alpha$ *and* $\pi\alpha \approx \alpha$ *for any* $\pi \in \mathrm{Aut}(M)$.
2. $\alpha \approx \beta$ *if and only id there exists some* $\pi \in \mathrm{Aut}(M)$ *such that* $\pi\alpha \sim \beta$.

Outer automorphism groups

From Ahlbrandt, Ziegler:

- The map $\lambda\colon \mathrm{Bi}(M) \to \mathrm{Aut}(G)$ given by $\lambda(\alpha)(g) = \alpha g^k \alpha^{-1}$ for $\alpha \in \mathrm{Bi}_k(M)$ (k is the dimension).
- This is the map on the domain D/E of α induced by g.

What is the reverse map $\mathrm{Aut}(G) \to \mathrm{Bi}(M)$?

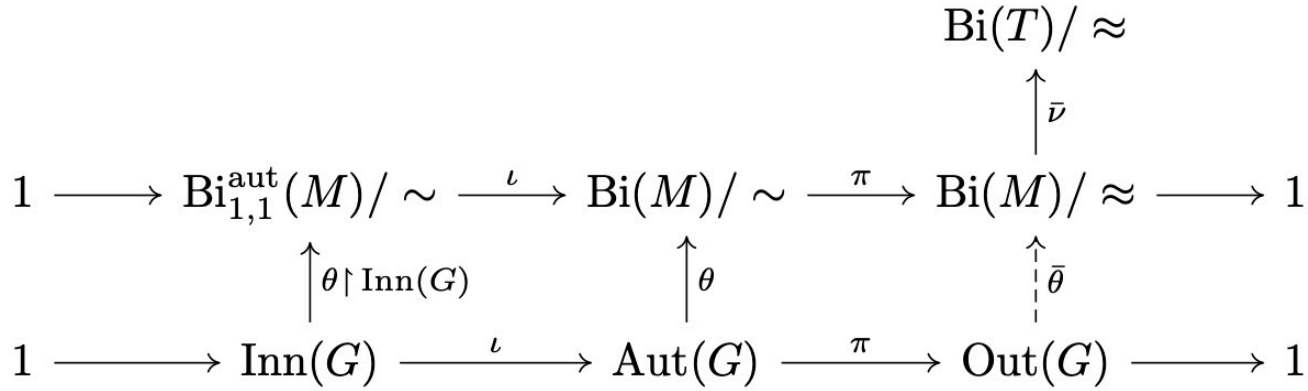
Lemma

Suppose that $p \in \mathrm{Aut}(G)$ and $k \in \mathbb{N}$. The following statements are equivalent for all $\bar{a}_i \in \mathbb{N}^k$ and $i < m$.

a. *There exists a k-dimensional bi-interpretation $\alpha \in \lambda^{-1}(p)$ such that $\alpha(\bar{a}_i) = b_i$ for all $i < m$.*
b. *$p(G_{\bar{a}_i}) \subseteq G_{b_i}$ for all $i < m$ and all $\bar{a}_i$ are in different orbits.*

Given $p \in \mathrm{Aut}(G)$, let $\bar{a}_i$, $\bar{b}_i$ as above and define $\theta(p) = \alpha$, where

$$\alpha(g^k \cdot \bar{a}_i) = p(g) \cdot b_i.$$


$\mathrm{Bi}(T)/\approx$
$\bar{\nu}$
$1 \longrightarrow \mathrm{Bi}^{\mathrm{aut}}_{1,1}(M)/\sim \xrightarrow{\iota} \mathrm{Bi}(M)/\sim \xrightarrow{\pi} \mathrm{Bi}(M)/\approx \longrightarrow 1$
$\theta\restriction \mathrm{Inn}(G)$
θ
$\bar{\theta}$
$1 \longrightarrow \mathrm{Inn}(G) \xrightarrow{\iota} \mathrm{Aut}(G) \xrightarrow{\pi} \mathrm{Out}(G) \longrightarrow 1$

Problems

Problem

Which groups arise aus outer automorphism groups of oligomorphic groups?

Problem (Nies, Paolini)

Is there an ω-categorical structure M with finite signature so that for $G = \mathrm{Aut}(M)$ we have an infinite $\mathrm{Out}(G)$?